

UPD: Evaluating LLMs as annotators

Deep Learning in NLP

(Groups 2 and 3, RISS annotation scheme)

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30 June 2026

Outline

1. Update on annotations
2. Evaluating LLM methods and models
 - Prompting

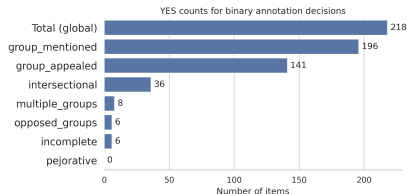
UPD Main Groups2-3: Human annotation throughput

Annotation statistics (23 June 2026 snapshot)

Parameter	Value
Available items	520
Annotated items	218
Triple+ annotated	124 (56.88 %)
Double-annotated	74 (33.94%)
Annotations	540
Annotators	7
Comments	4
Avg. min/item	1.32
Project start	2026-06-16
Project end	2026-06-23

gold standard: 98 sent_ids

Goal: 180 items across 3 weeks



	Alias	Items
group2	Horse	10
	Tiger	0
	Zebra	119
	Whale	37
group3	Robin	181
	Crane	0
	Raven	181
	lecturers	12

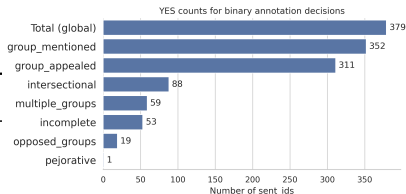
UPD Main Groups2-3: Human annotation throughput

Annotation statistics (30 June 2026 snapshot)

Parameter	Value
Available items	520
Annotated items	379
Triple+ annotated	272 (71.77 %)
Double-annotated	57 (15.04 %)
Annotations	980
Annotators	9
Comments	18
Avg. min/item	4.07
Project start	2026-06-16
Project end	2026-06-29 17:19

gold standard: 189 sent_ids

Goal: 180 items across 3 weeks



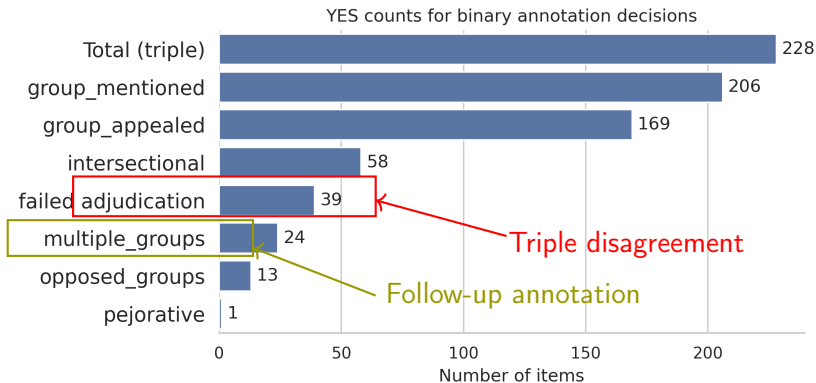
	Alias	Items
group2	Horse	10
	Tiger	14
	Zebra	181
	Whale	198
group3	Robin	181
	Crane	144
	Raven	240
	lecturers	12

Cross-annotations and adjudication

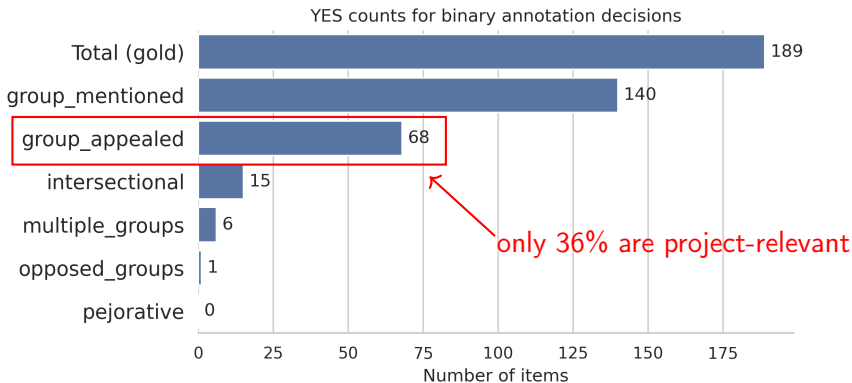
Every sent_id:

3 annotations → gold standard via adjudication

- Binary and multiclass variables: majority vote
- Tag lists: majority vote per tag
 - Exact-list match is too strict; tag union too permissive.
- No majority: adjudicate=True



Available adjudicated sent_ids (gold standard)



Disagreement range: 0 - 14

Mean: 1.91

Revised IAA approach

independent categories –» hierarchical logic

Type	Variable	<i>N</i>	<i>K</i>	Agree (%)	α
Binary	Group mentioned	228	2	72.81	0.484
	Group appealed	144	2	58.33	0.278
	Average	–	–	65.57	0.381
Binary	Intersectional	75	2	65.33	0.377
	Multiple groups	75	2	84.00	0.344
	Opposed groups	75	2	93.33	0.355
	Pejorative	75	2	98.67	0.000
Multiclass	Reasoning	75	4	41.33	0.386
	Polarity	75	3	85.33	0.577
	Stance	75	4	80.00	0.165
	Average	–	–	78.28	0.315
Overall		–	–	81.35	0.224

Do annotators agree that this category is present?

Category	N	K	Agree (%)	α
Age	75	2	68.00	0.146
Gender/sexuality	75	2	84.00	0.241
Family	75	2	78.67	0.162
Disability	75	2	84.00	0.176
Ethnicity	75	2	92.00	0.228
Migration	75	2	88.00	0.253
Citizenship	75	2	94.67	-0.005
Religion	75	2	97.33	0.223
Geography	75	2	86.67	0.206
Education	75	2	90.67	0.024
Income	75	2	76.00	0.132
Occupation	75	2	85.33	0.168
Non-RISS	75	2	85.33	0.006
Average	—	—	85.44	0.151

Per-tag IAA on social group types

How often do annotators agree on the presence of each group tag, regardless of other tags?

Krippendorff's α for social-group labels (frequency ≥ 5)

Category	Variable	Freq	N	Agree (%)	α
Age	School age	10	75	92.0	0.375
	Children	18	75	89.3	0.519
	Young adults	11	75	97.3	0.810
	Students	6	75	93.3	0.148
	Seniors	6	75	100.0	1.000
Gender	Women	26	75	98.7	0.957
	Men	7	75	98.7	0.853
Family	Parents	9	75	97.3	0.770
	Families	18	75	89.3	0.519
Disability	Disabled people	15	75	96.0	0.787
	Other health groups	12	75	90.7	0.387
Ethnicity	Named origin	6	75	93.3	0.148
	Ethnic minority	7	75	94.7	0.413

Category	Variable	Freq	<i>N</i>	Agree (%)	α
Migration	Refugees	17	75	97.3	0.873
	Immigrants	5	75	98.7	0.796
Geography	Rural	11	75	97.3	0.810
	Region-specific	6	75	93.3	0.148
Education	In education	6	75	92.0	-0.023
Income	Poor/low-income	6	75	96.0	0.489
	Working class	15	75	89.3	0.431
	Unemployed	6	75	97.3	0.659
Occupation	Small business owners	8	75	98.7	0.871
	Socio-cultural professionals	6	75	93.3	0.148
Other	Non-RISS	12	75	85.3	0.036
Average		10.4	–	94.6	0.538

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Experimental design

Open-weight models (instruction-tuned)

- Qwen3-1.7B⁶
- Qwen3.6-27B⁵
- gemma-4-31b-it²
- gpt-oss-120b⁴

Methods

- Zero-shot (using short definitions)
- Few-shot (using examples)
- Fine-tuning (not sure it is feasible)

LLM Annotation Quality based on LLM-to-gold IAA

Available gold data: 189 auto-adjudicated sent_ids

Meaning of usage limits in Colab

"You cannot currently connect to a GPU due to usage limits in Colab."

The restriction is based on **the duration** of GPU-allocated session.

The limit has to do with GPU-allocation limit after long sessions, not token budget.

Possible solution: Disconnect from the environment when not actively engaging with it.

Expected result: default formulation

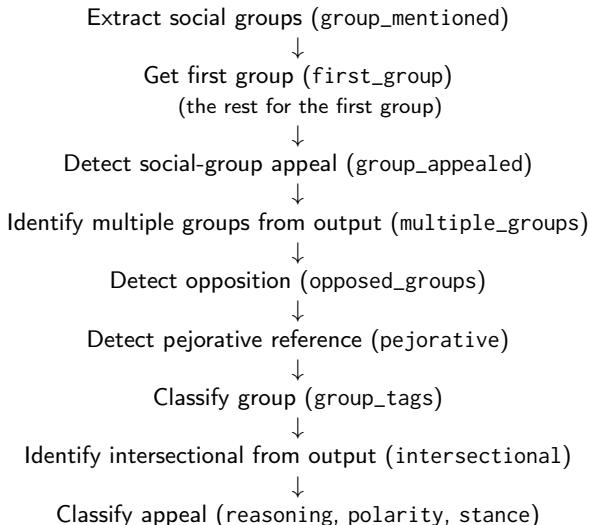
Draft 23 June 2026

Revised 30 June 2026

```
annotations = {  
  "groups": [],  
  "group_mentioned": "no",  
  "multiple_groups": "no",  
  "first_group": None,  
  "group_appealed": "no",  
  "opposed_groups": "no",  
  "pejorative": "no",  
  "group_tags": [],  
  "intersectional": None,  
  "reasoning": None,  
  "polarity": None,  
  "stance": None,  
}
```

```
annotations = {  
  "groups": [],  
  "group_mentioned": "no",  
  "multiple_groups": "NA",  
  "first_group": "",  
  "group_appealed": "NA",  
  "opposed_groups": "NA",  
  "pejorative": "NA",  
  "group_tags": ["NA"],  
  "intersectional": "NA",  
  "reasoning": "NA",  
  "polarity": "NA",  
  "stance": "NA",  
}
```

Revised hierarchical annotation pipeline



Pipeline terminates with defaults if no mention or no appeal is detected.

Project links

- Groups2-3 shared annotation project in *Deeplearning in NLP course* at Johannes's Label Studio workspace (reset and released to students on 15/06/2026, 23:57)
<https://app.heartex.com/projects?workspace=139506>
- github repository containing
 - raw output of Label Studio annotations
 - postprocessing script (transforming annotations to dataset, human-human IAA calculations, majority voting gold standard generation)
 - LLMs evaluation
 - preprocessed full data for Hansard 1980, 2025, 2026 (complete with speaker-party information)
 - a script to generate automatic annotations for Hansard data

<https://github.com/kunilovskaya/DLDH-SocSci>

Other definitions

`item == sent_id`

`label, tag, value ==` a choice made in the interface
(`no_group_mention`), or its internal code
(`AGE_CHILD`), or its formalized version
(`no_group_mention`, if selected $\rightarrow$
`group_mentioned`: “no”)

`incomplete` : `group_appealed` = yes but at least one of the
required appeal dimensions (reasoning, polarity, or
stance) was missing

`adjudicate` : annotator disagreement could not be resolved
automatically and requires manual review (e.g. triple
disagreement, a multi-label tag is not seen at least 3
times)

`disagreement` : disagreement score per item, the number of
annotation variables for which at least two annotators
provided different values. The higher, the more
contested the item.

1. Convert two hierarchical negatives into a formal 0/1 category

LS output		dataset	
group_appeal_detection	stance	group_mentioned	group_appealed
no group mentioned	–	no	no
NaN	endorse	yes	yes
no group appealed	–	yes	no

2. Disagreement score and auto-adjudication¹

binary	multiclass	multi-label
AAA → +0, adj=no	AAA → +0, adj=no	[A][A][B] → +1, adj=no
AAB → +1, adj=no	AAB → +1, adj=no	[A][B][C] → +2, adj=yes
ABB → +1, adj=no	ABC → +2, adj=yes	[A,B,C][A][D] +3, adj=no

¹For reasoning, polarity, stance, and group_tags, disagreement is only evaluated when gold[“group_appealed”] = yes. Range: [0,8]; mean: 2.92

References I

- [1] Artstein, R. & Poesio, M. (2008). Survey article: Inter-coder agreement for computational linguistics. Computational linguistics, 34(4), 555–596.
- [2] Google (2026). Gemma 4 Model Card.
- [3] Krippendorff, K. (2004). Content Analysis: An Introduction to Its Methodology. Thousand Oaks, CA: Sage Publications, 2 edition.
- [4] OpenAI (2025). gpt-oss-120b & gpt-oss-20b model card.
- [5] Qwen Team (2026). Qwen3.6-27B: Flagship-level coding in a 27B dense model.
- [6] Team, Q. (2025). Qwen3 technical report.